

Calculus I Lesson 21

1) $y = f(x) = (3x + 2)^4$

Find $y' = f'(x)$.

Let $u = 3x + 2$

a) $du / dx = \underline{\hspace{1cm}} ?$

b) Rewrite $y = (3x + 2)^4$ in terms of u : $y = \underline{\hspace{1cm}} ?$

c) $dy / du = \underline{\hspace{1cm}} ?$

d) Rewrite dy / du in terms of x . $dy / du = \underline{\hspace{1cm}} ?$

e) $y' = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \underline{\hspace{1cm}} ?$

(Note: Final answer must be in terms of x .)

2) $y = f(x) = (2x + 5)^3$

Find $y' = f'(x)$.

Let $u = 2x + 5$

a) $du / dx = \underline{\hspace{1cm}} ?$

b) Rewrite $y = (2x + 5)^3$ in terms of u : $y = \underline{\hspace{1cm}} ?$

c) $dy / du = \underline{\hspace{1cm}} ?$

d) Rewrite dy / du in terms of x . $dy / du = \underline{\hspace{1cm}} ?$

e) $y' = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \underline{\hspace{1cm}} ?$ (Note: Final answer must be in terms of x .)

3) $y = f(x) = (9x + 5)^{1/3}$

Find $y' = f'(x)$.

Let $u = 9x + 5$

a) $du / dx = \underline{\hspace{1cm}} ?$

b) Rewrite $y = (9x + 5)^{1/3}$ in terms of u : $y = \underline{\hspace{1cm}} ?$

c) $dy / du = \underline{\hspace{1cm}} ?$

d) Rewrite dy / du in terms of x . $dy / du = \underline{\hspace{1cm}} ?$

e) $y' = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \underline{\hspace{1cm}} ?$ (Note: Final answer must be in terms of x .)

4) $y = f(x) = \sqrt[3]{7x^2 + 12}$

Find $y' = f'(x)$.

Hint: $\sqrt[3]{7x^2 + 12} = (7x^2 + 12)^{1/3}$

Let $u = 7x^2 + 12$

a) $du / dx = \underline{\hspace{1cm}} ?$

b) Rewrite $y = (7x^2 + 12)^{1/3}$ in terms of u : $y = \underline{\hspace{1cm}} ?$

c) $dy / du = \underline{\hspace{1cm}} ?$

d) Rewrite dy / du in terms of x . $dy / du = \underline{\hspace{1cm}} ?$

e) $y' = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \underline{\hspace{1cm}} ?$ (Note: Final answer must be in terms of x .)

5) $y = f(x) = \frac{1}{x-5}$

Find $y' = f'(x)$.

Hint: $\frac{1}{x-5} = \frac{1}{(x-5)^1} = (x-5)^{-1}$

Let $u = x - 5$

a) $du/dx = \underline{\hspace{1cm}}?$

b) Rewrite $y = (x-5)^{-1}$ in terms of u : $y = \underline{\hspace{1cm}}?$

c) $dy/du = \underline{\hspace{1cm}}?$

d) Rewrite dy/du in terms of x . $dy/du = \underline{\hspace{1cm}}?$

e) $y' = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \underline{\hspace{1cm}}?$ (Note: Final answer must be in terms of x .)

6) $y = f(x) = \frac{1}{\sqrt{3x+1}}$

Find $y' = f'(x)$.

Hint: $\frac{1}{\sqrt{3x+1}} = \frac{1}{(3x+1)^{1/2}} = (3x+1)^{-1/2}$

Let $u = 3x + 1$

a) $du/dx = \underline{\hspace{1cm}}?$

b) Rewrite $y = (3x+1)^{-1/2}$ in terms of u : $y = \underline{\hspace{1cm}}?$

c) $dy/du = \underline{\hspace{1cm}}?$

d) Rewrite dy/du in terms of x . $dy/du = \underline{\hspace{1cm}}?$

e) $y' = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \underline{\hspace{1cm}}?$ (Note: Final answer must be in terms of x .)

7) $y = f(x) = \frac{1}{\sqrt[4]{2x-5}}$

Find $y' = f'(x)$.

Hint: $\frac{1}{\sqrt[4]{2x-5}} = \frac{1}{(2x-5)^{1/4}} = (2x-5)^{-1/4}$

Let $u = 2x - 5$

a) $du/dx = \underline{\hspace{1cm}}?$

b) Rewrite $y = (2x-5)^{-1/4}$ in terms of u : $y = \underline{\hspace{1cm}}?$

c) $dy/du = \underline{\hspace{1cm}}?$

d) Rewrite dy/du in terms of x . $dy/du = \underline{\hspace{1cm}}?$

e) $y' = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \underline{\hspace{1cm}}?$ (Note: Final answer must be in terms of x .)

